

Classification: LDA & QDA

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Overview

- Classification (Discriminant analysis) is a predictive task in which the response takes the values across several categories.
- Examples
 - Predicting whether a patient will develop breast cancer or remain healthy, given genetic information
 - Predicting whether or not a user will like a new product, based on user covariates and a history of his/her previous ratings
- In classification we learn from data where true classes are known.

X and Y
features and response.

General setup

- Labeled data: (Y_i, X_i) , $i = 1, \dots, n$, where values of Y_i represent class labels (categories 1 to K), and X_i are feature vectors
- Aim: For new observation x_0 , predict its class label.

Note: Class labels are unordered categories

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Bayes classifier

Bayes classifier $\arg\max_k P(Y=k | X=x_0)$

Assumption $P(Y=k) = \pi_k$

$P(X=x | Y=k) = f_k(x) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} e^{-\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)}$

K classes $\left\{ \begin{array}{l} \mathcal{N}(\mu_1, \Sigma_1) \\ \mathcal{N}(\mu_2, \Sigma_2) \\ \vdots \\ \mathcal{N}(\mu_K, \Sigma_K) \end{array} \right.$

$\Rightarrow \arg\max_k P(X=x_0 | Y=k) P(Y=k)$

- Find $P(Y | X)$. Then, given an input x_0 , we predict its class using the following rule:

$$C^{\text{bayes}}(x_0) = \arg\max_{k \in \{1, \dots, K\}} P(Y = k | X = x_0).$$

- Both LDA and QDA are based on Bayes classifier. They differ in assumptions about the data and how they estimate $P(Y | X)$

How to obtain Bayes classifier?

Strategy: Instead of estimating $P(Y | X)$ directly, we will estimate:

- $P(X | Y)$: Distribution of the features in each class
- $P(Y)$: Probability of each class

Given $P(X | Y)$ and $P(Y)$, we then use Bayes rule to obtain the estimate:

$$P(Y = k | X = x) = \frac{\overset{\text{argmax}_k}{P(X=x | Y=k)P(Y=k)}}{P(X=x)}$$

not important as it doesn't depend on k .

Note: The denominator is a normalization factor that is common to all classes. Therefore, it does not play any role in the comparison of posterior probabilities of any two classes.

Linear discriminant analysis (LDA)

Suppose that:

- We know $P(Y = k) = \pi_k$ exactly.
- $P(X = x \mid Y = k)$ is multivariate normal with density:

$$f_k(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1} (x - \mu_k)}$$

μ_k : Mean of the inputs for category k . Σ : Covariance matrix that is common to all categories.

Then, we can derive the Bayes classifier under the assumptions.

LDA

By Bayes rule, the probability of category k , given the input x is:

$$P(Y = k \mid X = x) = \frac{f_k(x)\pi_k}{P(X = x)}$$

The denominator does not depend on the response k , so we can write it as a constant:

$$P(Y = k \mid X = x) = C \times f_k(x)\pi_k$$

Plugging in the definition of a multivariate normal density for $f_k(x)$:

$$P(Y = k \mid X = x) = \frac{C\pi_k}{(2\pi)^{p/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T \Sigma^{-1}(x-\mu_k)}$$

LDA

Now, let us absorb everything that does not depend on k into a constant C' :

$$P(Y = k \mid X = x) = C' \pi_k e^{-\frac{1}{2}(x - \mu_k)^T \Sigma^{-1} (x - \mu_k)}$$

and take the logarithm of both sides:

$$\log P(Y = k \mid X = x) = \log C' + \log \pi_k - \frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) .$$

This constant is the same for every category, k . So we want to find the maximum of this over k .

LDA

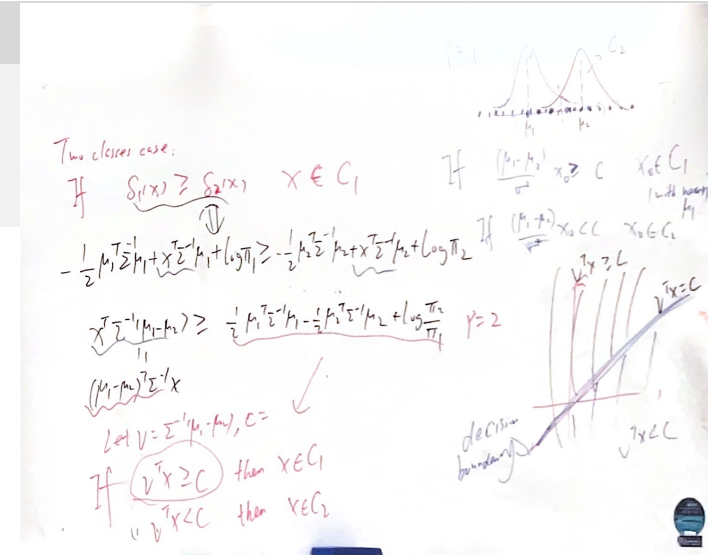
Goal, maximize the following over k :

$$\begin{aligned}
 & \log \pi_k - \frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) . \\
 &= \log \pi_k - \frac{1}{2} [x^T \Sigma^{-1} x + \mu_k^T \Sigma^{-1} \mu_k] + x^T \Sigma^{-1} \mu_k \\
 &= C'' + \log \pi_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + x^T \Sigma^{-1} \mu_k
 \end{aligned}$$

We define the objective:

$$\delta_k(x) = \log \pi_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + x^T \Sigma^{-1} \mu_k$$

At an input x , we predict the response with the highest $\delta_k(x)$.



LDA has linear decision boundary

What is the decision boundary? It is the set of points for which 2 classes are equally probable:

$$\underbrace{\log \pi_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k}_{\text{constant to } \vec{x}} + \underbrace{\vec{x}^T \Sigma^{-1} \mu_k}_{\text{constant to } \vec{x}} = \underbrace{\log \pi_\ell - \frac{1}{2} \mu_\ell^T \Sigma^{-1} \mu_\ell}_{\text{constant to } \vec{x}} + \underbrace{\vec{x}^T \Sigma^{-1} \mu_\ell}_{\text{constant to } \vec{x}}$$

This is a linear equation in $\vec{x}$!

- Two classes case: Assign $\vec{x}$ to class 1 if $\delta_1(\vec{x}) - \delta_2(\vec{x}) > 0$ or equivalently

$$(\mu_1 - \mu_2)^T \Sigma^{-1} \vec{x} > c, \quad \text{in one dimension, case.}$$

$$V = \Sigma^{-1}(\mu_1 - \mu_2)$$

$$\frac{\mu_1 - \mu_2}{\sigma^2} \vec{x} > c$$

where

$$c = \frac{1}{2} (\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 + \mu_2) + \log \left(\frac{\pi_2}{\pi_1} \right)$$

If $V^T \vec{x} \geq c$ then $\vec{x} \in C_1$
 else then $\vec{x} \in C_2$

LDA has linear decision boundary

The feature space is divided into two decision regions if there are two classes:

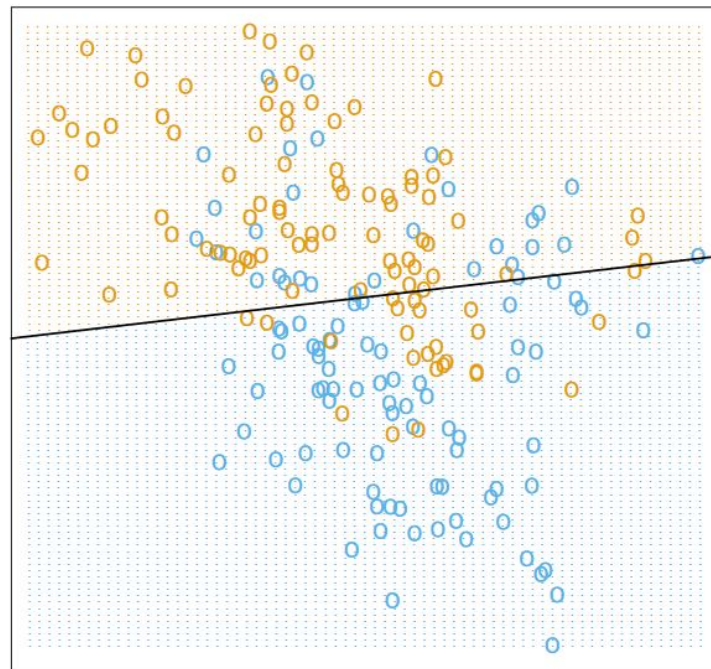


Figure: LDA performs classification based on decision regions separated by linear boundaries

LDA from data

To use $\delta_k(x)$ in practice, we need to estimate μ_k and Σ (and also π_k if unknown) from the **training data**. For class k , let

- n_k = total number of observations
- $\bar{x}_k$ and S_k = sample mean and covariance matrix of predictor values

Estimates:

$$\hat{\pi}_k = \frac{n_k}{n}, \quad \hat{\mu}_k = \bar{x}_k, \quad \hat{\Sigma} = \frac{(n_1 - 1) S_1 + (n_2 - 1) S_2 + \cdots + (n_K - 1) S_K}{n - K}$$

- “Natural” estimators
- $\hat{\Sigma}$ is the pooled sample covariance matrix
- Plug in to get:

$$\hat{\delta}_k(x) = \hat{\mu}_k^T \hat{\Sigma}^{-1} x - \frac{1}{2} \hat{\mu}_k^T \hat{\Sigma}^{-1} \hat{\mu}_k + \log(\hat{\pi}_k)$$

Two classes case

Assign $\mathbf{x}$ to class 1 if $\hat{\delta}_1(\mathbf{x}) - \hat{\delta}_2(\mathbf{x}) > 0$ or equivalently (verify)

$$(\hat{\mu}_1 - \hat{\mu}_2)^T \hat{\Sigma}^{-1} \mathbf{x} > \hat{c},$$

where

$$\hat{c} = \frac{1}{2} (\hat{\mu}_1 - \hat{\mu}_2)^T \hat{\Sigma}^{-1} (\hat{\mu}_1 + \hat{\mu}_2) + \log \left(\frac{\hat{\pi}_2}{\hat{\pi}_1} \right)$$

Three classes case

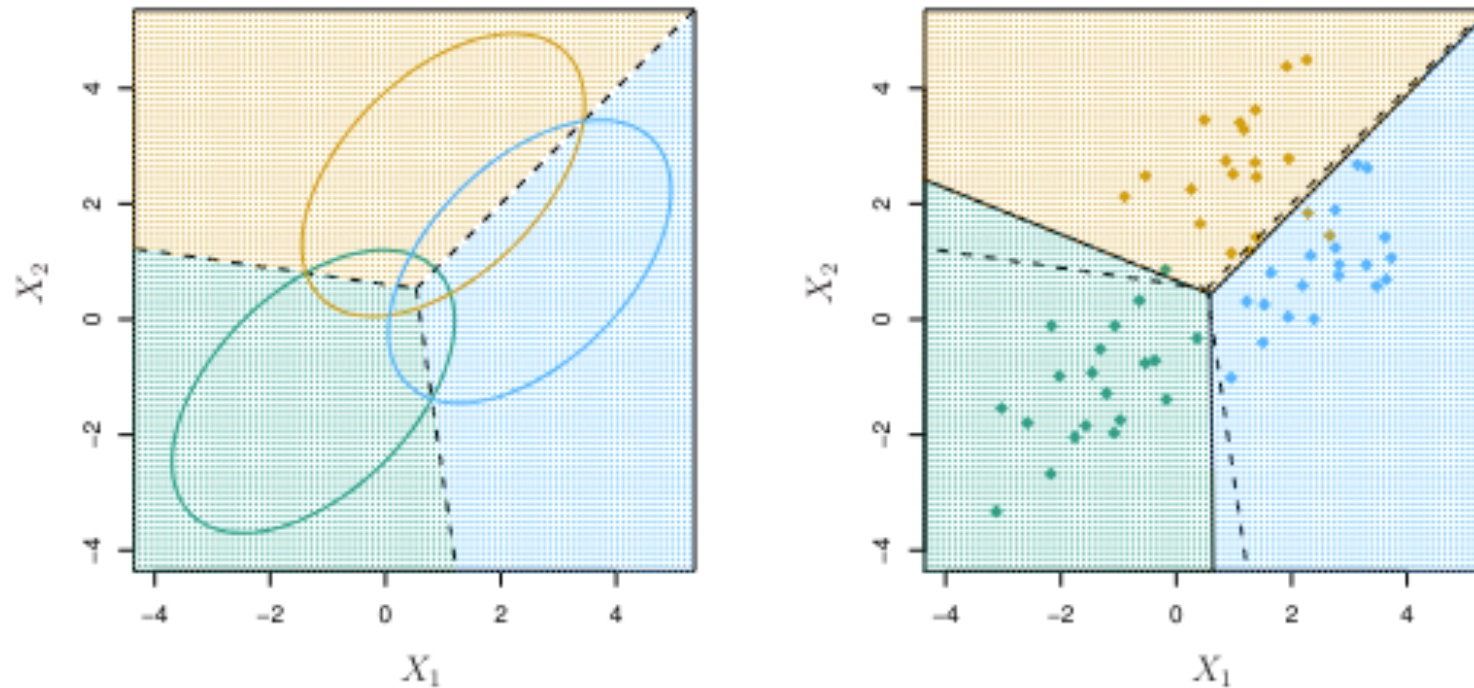


Figure: An example with three classes. The observations from each class are drawn from a multivariate Gaussian distribution with $p = 2$, with a class-specific mean vector and a common covariance matrix. Left: Ellipses that contain 95% of the probability for each of the three classes are shown. The dashed lines are the Bayes decision boundaries. Right: 20 observations were generated from each class. The Bayes decision boundaries are once again shown as dashed lines, and the corresponding LDA decision boundaries are indicated using solid black lines.

Quadratic discriminant analysis

- The assumption that the inputs of every class have the same covariance Σ can be quite restrictive.
- Given an input, it is easy to derive an objective function:

$$\delta_k(x) = \log \pi_k - \frac{1}{2} \mu_k^T \Sigma_k^{-1} \mu_k + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} x^T \Sigma_k^{-1} x - \frac{1}{2} \log |\Sigma_k|$$

This objective is now quadratic in x .

- In quadratic discriminant analysis we estimate a mean $\hat{\mu}_k$ and a covariance matrix $\hat{\Sigma}_k$ for each class separately.

An illustration of QDA with two classes

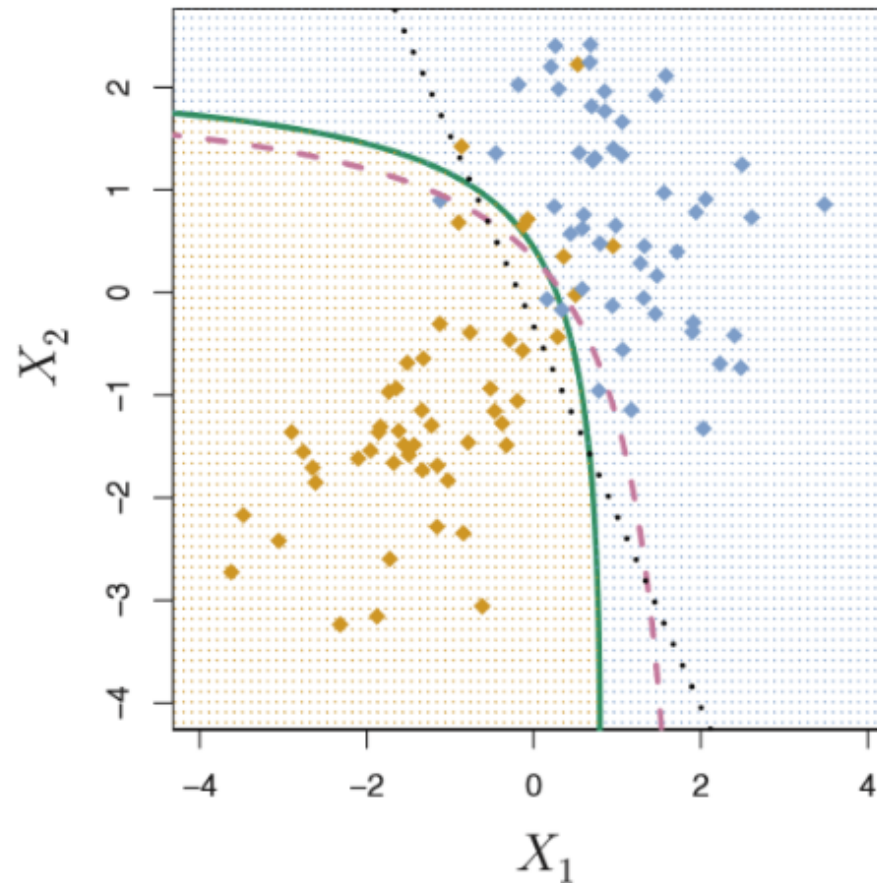


Figure: Bayes decision boundary (purple dashed curve), LDA boundary (black dotted line), QDA boundary (green solid curve)

LDA vs. QDA

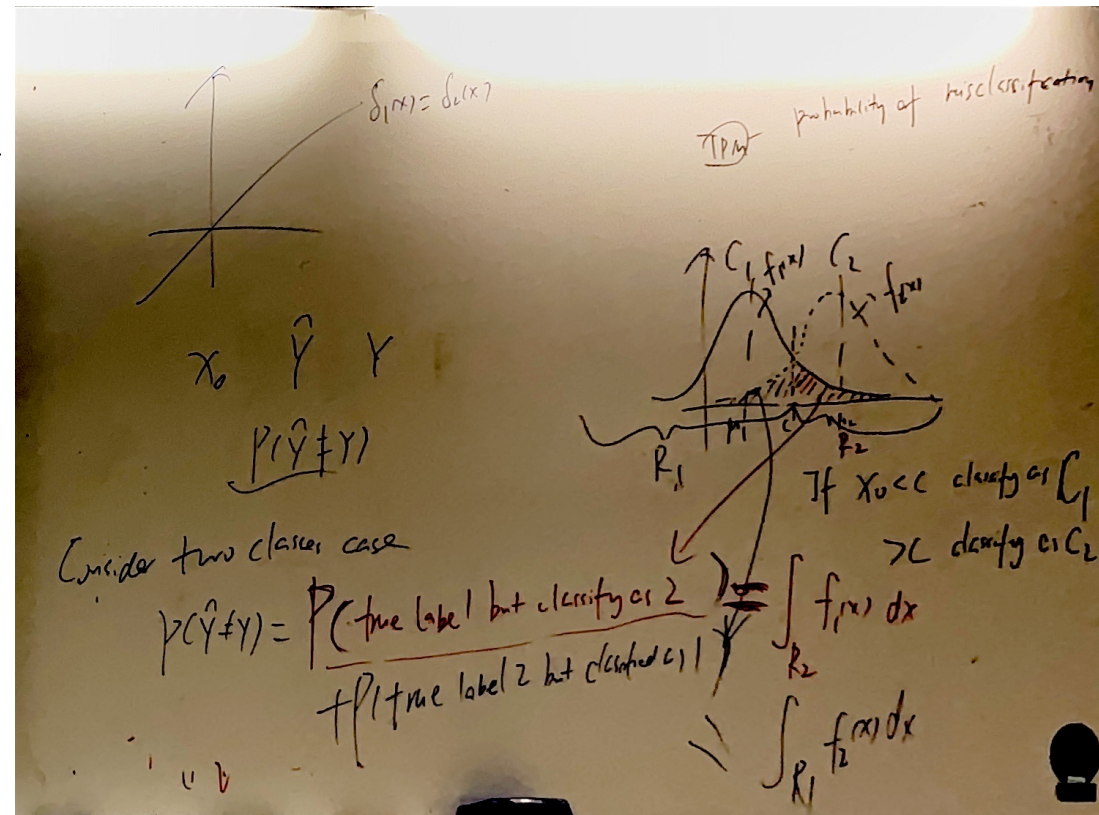
- Both LDA and QDA are Bayes classifiers

$$\hat{y}_0 = \operatorname{argmax}_y P(Y = y | X = x_0)$$

- Key Assumptions: Both assume $P(X|Y)$ follows Normal distribution; and both estimate prior probabilities $P(Y)$ from data
- Difference:
 - **LDA**: Assumes equal covariance matrices
→ Results in linear decision boundaries
 - **QDA**: Allows different covariance matrices
→ Results in quadratic decision boundaries

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TPM

Let $\hat{Y}$ be the predicted label and Y the true label.

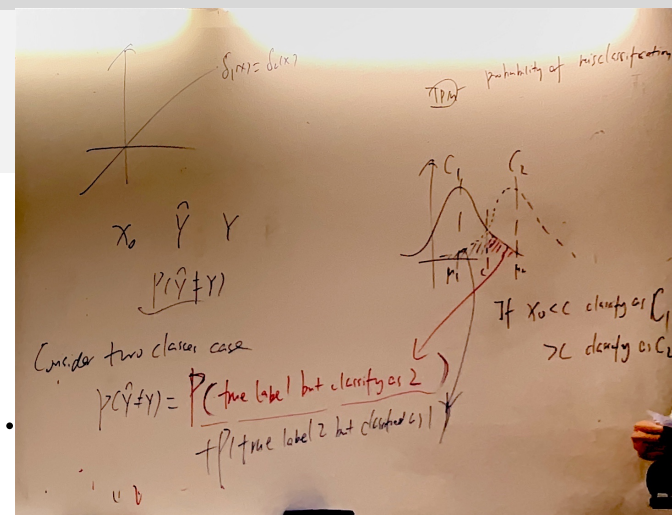
- Error: Zero-one error, i.e.,

$$I(\hat{Y} \neq Y) = \begin{cases} 1, & \text{if } \hat{Y} \neq Y \\ 0, & \text{if } \hat{Y} = Y \end{cases}$$

- Total probability of misclassification (TPM):

$$E\{I(\hat{Y} \neq Y)\} = P(\hat{Y} \neq Y)$$

- TPM can be referred to as “expected error rate”, and can be understood as $\frac{\text{Number of wrong predictions}}{\text{Total number of predictions}}$



If there are two classes, the “total probability of misclassification” is

$$TPM = \pi_1 P(2 \mid 1) + \pi_2 P(1 \mid 2),$$

where $P(k \mid i) = P(\text{classifying item as } k \mid \text{true label is } i)$.

Now we aim to derive the TPM for “theoretical LDA”. Consider $\pi_1 = \pi_2 = \frac{1}{2}$ and $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ are multivariate normal with the same covariance Σ but different mean μ_1, μ_2 .

TPM for LDA (two balanced classes case)

Put $Y = (\mu_1 - \mu_2)^T \Sigma^{-1} \mathbf{x}$. This is normal, variance $(\mu_1 - \mu_2)^T \Sigma^{-1} (\mu_1 - \mu_2) \stackrel{\text{def}}{=} \Delta^2$ and mean

$$\nu_1 = E[Y \mid 1] = (\mu_1 - \mu_2)^T \Sigma^{-1} \mu_1,$$

$$\nu_2 = E[Y \mid 2] = (\mu_1 - \mu_2)^T \Sigma^{-1} \mu_2,$$

with $\nu_1 - \nu_2 = \Delta^2$. We have

$$P(1 \mid 2) = P\left(Y \geq \frac{\nu_1 + \nu_2}{2} \mid 2\right) = P\left(\frac{Y - \nu_2}{\Delta} \geq \frac{\Delta}{2}\right) = \Phi\left(-\frac{\Delta}{2}\right).$$

Similarly $P(2 \mid 1) = \Phi(-\Delta/2)$ and so

$$TPM = \Phi\left(-\frac{\Delta}{2}\right).$$

Indeed, this is the minimum of TPM that can be achieved by any classifier when we have two balanced classes that follow normal distributions with the same covariance.

Note that for small Δ this is almost 0.5.

Bayes classifier is optimal

Theorem

Bayes classifier is optimal in the sense that it minimizes the TPM.

The actual performance of LDA and QDA depends on how well their assumptions match reality.

Estimate error rate

- The minimum of TPM can be calculated when the population density functions are known, but in practice usually not known.
- Confusion matrix:

		Predicted class		
		- or Null	+ or Non-null	Total
True class	- or Null	True Neg. (TN)	False Pos. (FP)	N
	+ or Non-null	False Neg. (FN)	True Pos. (TP)	P
	Total	N*	P*	

Estimate the error rate

- Apply the classifier to the training sample and compute the fraction of items misclassified? Usually underestimate the true error.
- If the sample size is large, split the total sample into a training sample and a validation sample. The training sample is used to construct the classification function, and the validation sample is used to evaluate it. The error rate is determined by the proportion misclassified in the validation sample.
- If sample size not large, can use Leave-One-Out method (LOO): Omit one observation from the training sample; develop the classifier from the remainder of the sample; classify the observation which was held out. Repeat n times; estimate the error rate from the resulting confusion matrix.

Some practical consideration

In real applications, different types of misclassification may have different costs. Misclassifying class 1 as 2 may be more dangerous than misclassifying class 2 as 1.

Example. Predicting default

Used LDA to predict credit card default in a dataset of 10 K people.
 Predicted “yes” if $\hat{P}(\text{default} = \text{yes} \mid \mathbf{X}) > 0.5$. Use LOO method:

		True default status		
		No	Yes	Total
Predicted default status	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- The error rate among people who do not default (false positive rate) is very low.
- However, the error rate among people who do default (false negative rate) is 76% !
- False negatives may be a bigger source of concern!
- One possible solution: Change the threshold.

Example. Predicting default

Changing the threshold to 0.2 makes it easier to classify to “yes”.
 Predicted “yes” if $\hat{P}(\text{def ault} = \text{yes} \mid X) > 0.2$.

		True default status		
		No	Yes	Total
Predicted default status	No	9,432	138	9,570
	Yes	235	195	430
	Total	9,667	333	10,000

Note that the rate of false positives became higher. That is the price to pay for fewer false negatives.

$$TPM = \underbrace{\int_{R_2} f_1(x) \pi_1 dx}_{\text{probability misclassifying } C_1 \text{ to } C_2} + \int_{R_1} f_2(x) \pi_2 dx \quad \begin{matrix} C_1 \text{ to } C_1 \\ C_1 \text{ to } C_2 \end{matrix}$$

$$P(Y=1|X=x) > 0.5 \quad x \in C_1$$

$$\Leftrightarrow P(Y=1|X=x) > P(Y=2|X=x) \quad x \in C_1$$

$$\left(\Leftrightarrow \pi_1 f_1(x) > \pi_2 f_2(x) \quad x \in C_1 \right)$$

$$TPM = \underbrace{\int_{R_2} f_1(x) \pi_1 dx}_{\text{Prob of misclassifying } C_1 \text{ as } C_2} + \underbrace{\int_{R_1} f_2(x) \pi_2 dx}_{C_2 \text{ as } C_1}$$

$$\Rightarrow \left. \begin{aligned} &R_1: \left\{ \begin{aligned} &f_1(x) \pi_1 > f_2(x) \pi_2 \\ &\frac{f_1(x)}{f_2(x)} > \frac{\pi_2}{\pi_1} \end{aligned} \right\} \end{aligned} \right\}$$

$$\int_{R_2} C(2/1) f_1(x) \pi_1 dx + \int_{R_1} C(1/2) f_2(x) \pi_2 dx \rightarrow$$

$$\left(C(2/1) f_1(x) \pi_1 > C(1/2) f_2(x) \pi_2 \right)$$

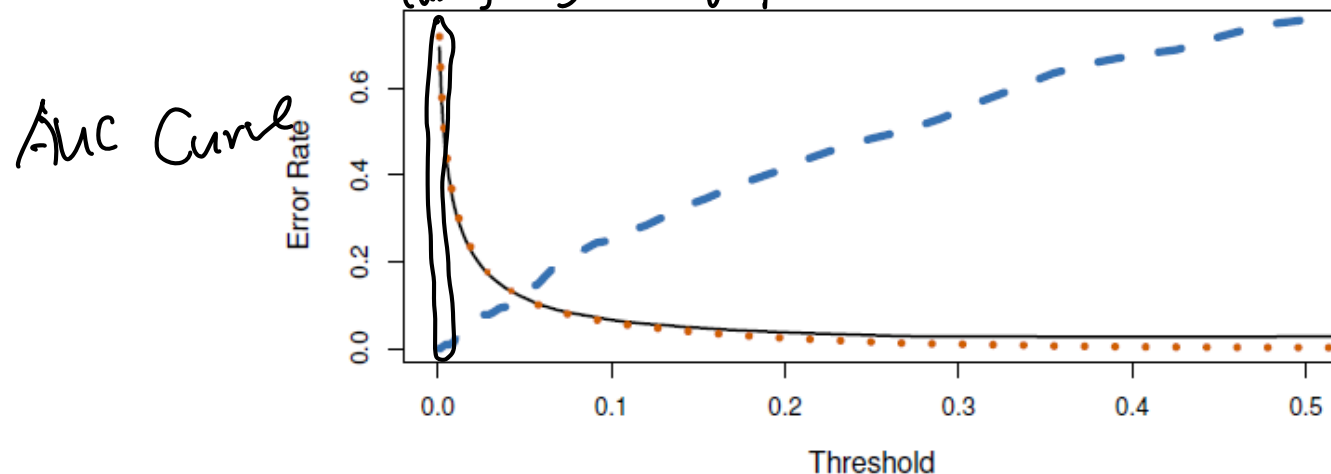
$$\frac{f_1(x)}{f_2(x)} > \frac{\pi_2 C(1/2)}{\pi_1 C(2/1)}$$

Example. Predicting default

Let's visualize the dependence of the error on the threshold:

error = 1 and error = 0 depends on threshold condition

classify every to one group



- ▶ — — — False negative rate (error for defaulting customers)
- ▶ False positive rate (error for non-defaulting customers)
- ▶ — 0-1 loss or total error rate.

Changing classification threshold to trade off between false positives and false negatives is an example of tuning a classifier to perform well with respect to problem-specific objective.

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Fisher's discriminant rule

Ronald Fisher's idea: Transform the multivariate observations $\mathbf{x}$ to univariate observations y such that the y 's derived from populations π_1 and π_2 were separated as much as possible.

We consider the two-classes case with common population to illustrate the idea.

$$\mathbf{w} = \Sigma^{-1}(\mu_1 - \mu_2)$$

$$\mathbf{w}^T \mathbf{x} > c \quad \mathbf{x} \in C_1$$

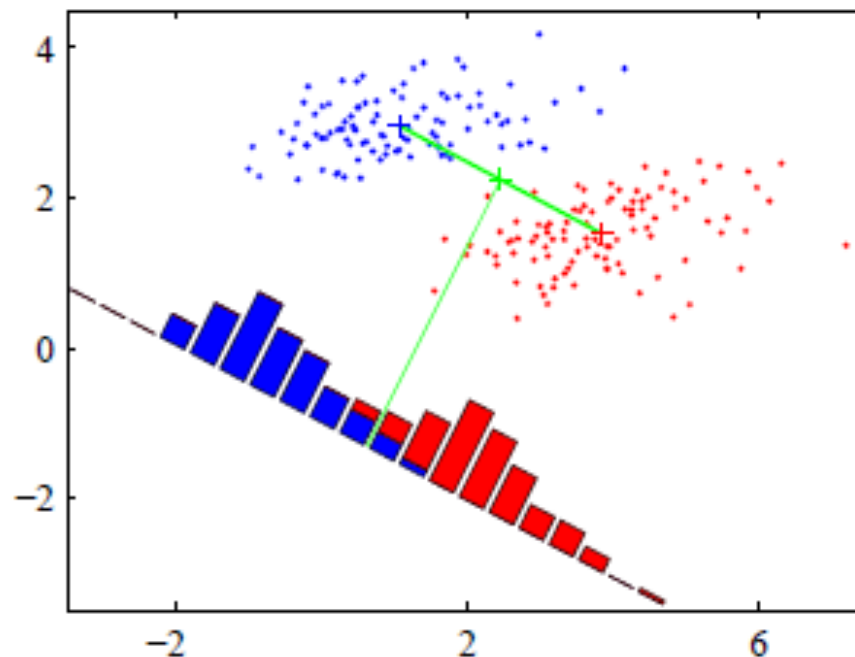
$$\mathbf{w}^T \mathbf{x} < c \quad \mathbf{x} \in C_2$$

$$\text{If } \|\mathbf{w}\| = 1$$

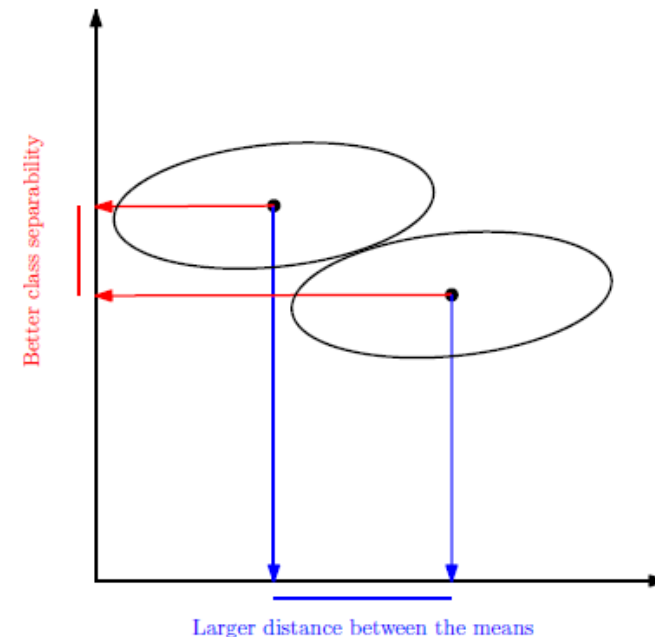
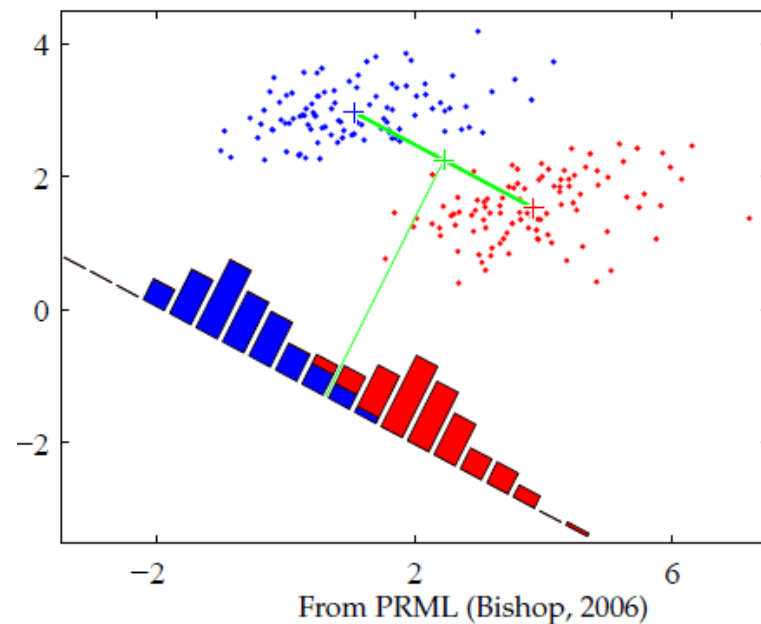
$$\mathbf{x}^T \mathbf{w} = \|\mathbf{x}\| \|\mathbf{w}\| \cos \theta$$

$$= \|\mathbf{x}\| \cos \theta$$

$$\Leftrightarrow \mathbf{w}^T \mathbf{x} > c \quad \frac{c}{\cos \theta} > \|\mathbf{x}\|$$



Fisher's discriminant rule



Separate samples of distinct groups by projecting them onto a space that

- Maximizes their between-class separability while
- Minimizing their within-class variability

- Aim: Find the projection that maximizes the class separation.
- Consider two classes: C_1 with n_1 points and C_2 with n_2 points -
Corresponding mean vectors:

$$\hat{\mu}_1 = \frac{1}{n_1} \sum_{i \in C_1} x_i, \quad \hat{\mu}_2 = \frac{1}{n_2} \sum_{i \in C_2} x_i$$

- The separation can be quantified by the difference between the projected class means expressed in standard deviation units:
Projected mean difference is given as

$$m_2 - m_1 = a^\top \hat{\mu}_2 - a^\top \hat{\mu}_1 = a^\top (\hat{\mu}_2 - \hat{\mu}_1)$$

Within-class variation is given as

$$s^2 = \frac{1}{n_1 + n_2 - 2} \sum_{k=1}^2 \sum_{i \in C_k} \left(a^\top x_i - a^\top \hat{\mu}_k \right)^2,$$

Fisher criterion

Maximize the Fisher criterion:

$$\frac{\text{Between-class separation}}{\text{Within-class variability}} = \frac{(m_2 - m_1)^2}{s^2} = \frac{[a^\top (\hat{\mu}_2 - \hat{\mu}_1)]^2}{a^\top S_{pooled} a}$$

where

$$S_{pooled} = \frac{(n_1 - 1)S_1 + (n_2 - 1)S_2}{n_1 + n_2 - 2}$$

Use the result on page 22 of the slides "Chapter 4", the maximal value for the Fisher criterion is attained at $a = (\hat{\mu}_1 - \hat{\mu}_2)^\top S_{pooled}^{-1}$ and the maximum is $(\hat{\mu}_1 - \hat{\mu}_2)^\top S_{pooled}^{-1} (\hat{\mu}_1 - \hat{\mu}_2)$.

Classifier based on Fisher's criterion

- Allocate $\mathbf{x}$ to π_1 if

$$(\hat{\mu}_1 - \hat{\mu}_2) S_{pooled}^{-1} \mathbf{x} \geq \frac{1}{2} (\hat{\mu}_1 - \hat{\mu}_2) S_{pooled}^{-1} (\hat{\mu}_1 + \hat{\mu}_2)$$

Allocate $\mathbf{x}_0$ to π_2 if the above inequality holds in the other direction.

This is the same as the classifier obtained using Bayes rule when there are two classes, the two populations are normal with equal covariance, and $\pi_1 = \pi_2$.

- The idea can be extended to cases involving several classes by employing the definition of between-class covariance defined in MANOVA, not discussed in detail here.

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Classification algorithms

LDA/QDA, logistic regression, K-nearest neighborhoods, Support vector machine, decision trees, neural networks...

Recap: Logistic regression

- Logistic regression can be used to predict the probability (odds) of a binary outcome.
- We model the joint probability as:

$$P(Y = 1 \mid X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}},$$
$$P(Y = 0 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}.$$

This is the same as using a linear model for the log odds:

$$\log \left[\frac{P(Y = 1 \mid X)}{P(Y = 0 \mid X)} \right] = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p.$$

LDA vs logistic regression

- Logistic regression is also a classification technique. (e.g., Assign X^* to class $Y=1$ if $\hat{\beta}_0 + \hat{\beta}_1 X_1^* + \cdots, \hat{\beta}_p X_p^* > 0$, because this means $P(Y = 1|X^*) > P(Y = 0|X^*)$).
- The difference between logistic regression and LDA is that the linear logistic model only specifies the conditional distribution $P(Y = k | X = x)$. No assumption is made about $P(X)$; while the LDA model specifies the joint distribution of X and Y . where ϕ is the Gaussian density function.
- Moreover, linear logistic regression is solved by maximizing the conditional likelihood of Y given X : $P(Y = k | X = x)$; while LDA maximizes the joint likelihood of Y and X : $P(X = x, Y = k)$.

LDA vs logistic regression

- If the additional assumption made by LDA is appropriate, LDA tends to be more accurate, since Bayes classifier is optimal.
- On the other hand, LDA is not robust to outliers. Because logistic regression relies on fewer assumptions, it seems to be more robust to the nonGaussian type of data.

LDA vs PCA

- LDA is also closely related to principal component analysis (PCA) in that they both look for linear combinations of variables which best explain the data.
- LDA explicitly attempts to model the difference between the classes of data. PCA, in contrast, does not take into account any difference in class.

